

ELEC5305 Project Proposal

Animal Sound Classification Using CNN and Data Augmentation

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1 Project Overview

Animal sounds contain useful information that can be used to identify different types of animals. For example, dogs, cats, cows, frogs, and birds produce very different sounds. Automatically recognising these sounds is a type of nonspeech sound recognition and can have possible applications in wildlife monitoring, animal monitoring, and automatic audio analysis.

The aim of this project is to build a simple animal sound classifier using a Convolutional Neural Network (CNN). The audio recordings will first be converted into Mel-spectrograms. These representations will then be used as input to the CNN to classify different types of animal sounds.

The project will also investigate whether adding noise to the training data can improve the performance of the CNN.

Research Question

Can noise augmentation improve the accuracy and robustness of a CNN for animal sound classification?

2 Background and Motivation

Environmental sound classification is the task of automatically recognising different sounds in an audio recording. Unlike speech recognition, the aim is not to recognise spoken words. Instead, the system can recognise other sounds such as animals, vehicles, rain, alarms, and other environmental events.

Piczak introduced the ESC-50 dataset for environmental sound classification [1]. It contains many different types of environmental sounds and has been widely used for testing audio classification methods. One group in the dataset contains animal sounds such as dog, cat, cow, pig, frog, sheep, and crow.

Convolutional Neural Networks have also been used for sound classification. A sound signal can be converted into a spectrogram, which shows the frequency content of the sound over time. A CNN can then learn patterns from this time-frequency representation and use these patterns for classification.

One possible problem is that a CNN may learn the training examples very well but perform less well on new or noisy recordings. Data augmentation is one possible method for reducing

this problem. It creates different versions of the training data while keeping the original class label. For audio data, examples include adding noise, changing the time position, or slightly changing the pitch.

Previous research has shown that data augmentation can help CNN-based animal audio classification [2]. More recent work has also shown that augmentation can improve acoustic classification when the amount of original training data is limited [3].

This project will study a simple version of this problem. Only noise augmentation will be investigated so that the project remains focused and the effect of the modification can be clearly evaluated.

3 Proposed Methodology

Dataset

The ESC-50 open dataset will be used for this project. Instead of using all 50 environmental sound classes, only the animal sound category will be selected.

The selected classes are:

- Dog
- Rooster
- Pig
- Cow
- Frog
- Cat
- Hen
- Insects
- Sheep
- Crow

Each class contains 40 recordings. Therefore, approximately 400 audio recordings will be used in this project.

ESC-50 already provides five predefined data folds. These folds will be used to separate the training, validation, and test data. For the initial experiment, three folds will be used for training, one fold for validation, and one fold for testing. The same data split will be used for both the baseline and modified models.

Audio Processing

The audio files will first be loaded and converted to the same audio format if required. Each audio recording will then be converted into a Mel-spectrogram.

The main processing sequence is:

$$\text{Audio} \rightarrow \text{Mel-spectrogram} \rightarrow \text{CNN} \rightarrow \text{Animal Class}$$

The Mel-spectrogram gives the CNN information about how the frequency content of an animal sound changes with time.

Baseline CNN

A simple CNN will first be trained using the original animal audio recordings. The model will contain convolutional layers, activation layers, pooling layers, and a final classification layer.

This model will be the baseline model.

Baseline: Original Training Audio + CNN

Its classification accuracy will first be measured on the test data.

Noise Augmentation

The second model will use exactly the same CNN structure. However, random noise will be added to some of the training audio recordings.

The animal label will not change. For example, a dog recording with added noise will still be labelled as “dog”.

The second training process will therefore be:

Modified Model: Original Audio + Noise Augmentation + Same CNN

Only the training data will be augmented. The original test data will not be used during training.

Python will be used for the project. Possible libraries include PyTorch, librosa, NumPy, scikit-learn, and Matplotlib.

4 Evaluation

The performance of the proposed animal sound classification system will be evaluated using a separate test dataset that is not used during training.

The main evaluation will focus on classification performance. The baseline CNN and the CNN trained with data augmentation will be tested under the same conditions so that their results can be compared fairly.

Classification accuracy will be used as the main measure of performance. Other evaluation methods may also be included if they are useful for understanding the behaviour of the models.

The results will be used to investigate whether data augmentation provides a useful improvement to the animal sound classification system.

5 Expected Outcomes

The expected outcome of this project is a working audio classification system that can recognise several types of animal sounds.

The project will explore the use of a CNN for animal sound classification and investigate the effect of data augmentation on the classification performance.

The final results are expected to provide a better understanding of whether data augmentation is useful for this task. The project will also produce code and documentation that describe the implementation and experiments.

6 Timeline

Weeks	Task
3–5	Literature review and download ESC-50 dataset
6	Prepare animal audio data and Mel-spectrograms
7–8	Build and train the baseline CNN
9	Evaluate the baseline model
10	Add noise augmentation and train the modified CNN
11	Compare accuracy and confusion matrices
12	Complete experiments and analyse results
13	Final report, video, and GitHub documentation

7 References

References

- [1] K. J. Piczak, “ESC: Dataset for Environmental Sound Classification,” *Proceedings of the 23rd ACM International Conference on Multimedia*, pp. 1015–1018, 2015. doi: 10.1145/2733373.2806390.
- [2] L. Nanni, G. Maguolo, and M. Paci, “Data Augmentation Approaches for Improving Animal Audio Classification,” *Ecological Informatics*, vol. 57, article 101084, 2020. doi: 10.1016/j.ecoinf.2020.101084.
- [3] J. MacIsaac, S. Newson, A. Ashton-Butt, H. Pearce, and B. Milner, “Improving Acoustic Species Identification Using Data Augmentation Within a Deep Learning Framework,” *Ecological Informatics*, vol. 83, article 102851, 2024. doi: 10.1016/j.ecoinf.2024.102851.